Design project report

02862010

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Introduction

Hearing loss is a result of many factors over one's lifetime. It could be genetic, age induced, caused by nervous system trauma, and much more[1]. In the UK, one out of six people have some form of hearing impairment [1]. In fact, according to RNID, the Royal National Institute for Deaf People, there are over 11 million people exhibiting symptoms of hearing loss [2]. However, just because these people suffer from hearing loss, does not mean they're not significant, functioning members of society. People with hearing loss still go to work, raise families, and have fun like everyone else. But in order to do it all, they first need to wake up on time. To combat the daily struggles deaf members of society encounter, we designed a new, enjoyable and efficient method of getting their day started.

Our new alarm clock for deaf people consists of an 8x8x8 LED cube that displays a series of captivating light patterns. Attached to the cube is a fan that blows a breeze onto the user to assist the wake-up process, digital clock., and off sensor. Deaf-friendly alarm clocks, however, are not a new concept. Other variations of the idea have been produced and sold in the past. Some examples include vibrating pillows, electric shock wristbands, and extra loud alarms [3]. All these products however lose out to the LED cube because they abruptly wake the user. The cube ensures a soothing waking process by providing a gentle yet effective amount of light to peacefully wake the user, as well as a light, yet concentrated fan breeze.

Design:

The construction of the light cube was a long, tedious process. Using a total of five hundred and twelve LEDs, we soldered together eight, 8x8 square-shaped sheets to form the final 3D cube. Each LED had two wires jutting out from the bottom.

Image 1. LED used in cube design.

Image 1 shows the LEDs the longer of the two was the cathode, which carries the negative current from the circuit; and the positively charged anode. Yet before we even began to solder the sheets, we needed to bend the anode of each LED to be able to solder them together horizontally. This is shown in image 2.

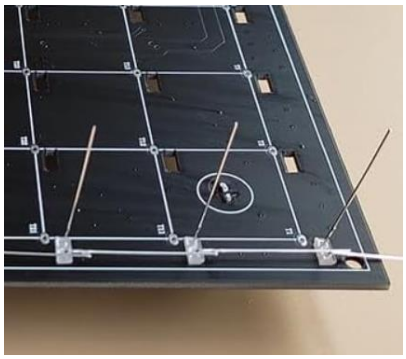

Image 2. Cube template and bent anode.

The soldering process was a huge learning experience. In our first attempt to solder an 8x8 sheet, we placed all sixty-four LEDs for the pane in our soldering template shown above. This template also doubled as the circuit board we bought to power the cube.

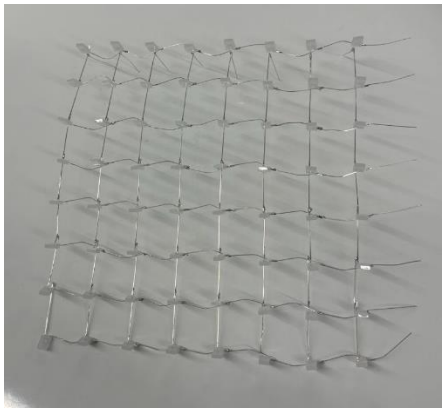

Image 3. First soldering attempt.

The first pane we constructed as shown in image 3 however, wasn't perfect. This first sheet was crooked which would make the vertical stacking of the sheets extremely difficult as the cathodes wouldn't line up. After this failure, it was clear that a new method of constructing the LED sheets was necessary. Through hours of brainstorming possible designs, we settled on implementing what we coined as the "Curly-Q" method.

Image 4. Curly-Q method.

The Curly-Q method required both the anode and cathode to be bent into tiny hoops (Qs) shown in image 4, perpendicular to the template. The anode Q was then bent so the opening of the hoop was horizontal to the template. This was done to allow us to feed wires through a line of 8 cathodes and solder each wire to the Q to form the induvial eight 8x8 sheets, and then stack the sheets on top of one another by inserting wires through the horizontally bent cathodes. The vertically aligned wire was also soldered to the horizontal cathode Qs. Once all the sheets were finished, we attempted to stack the sheets together by running wires through the all the cathodes and adding each sheet on top of the other. Once all the sheets were soldered onto the board, we would then begin to feed more wire through the cathode Qs and solder them together. Once the sheets were secured on the board, we

found once again that once again that our sheets were slightly misaligned, making it extremely difficult to run a rigid aluminum wire through a line of wonky LEDs.

Image 5. Feeding malleable aluminum wire through Curly Qs

To combat this, we bought a much more malleable wire shown in image 5, that could be bent through each cathode in order to accommodate the crooked Q's. The curly Q method was superior to our original method for two reasons, the most important being that it allowed the cathode and anode to never touch and that it allowed for the cube to have a strong skeleton due to the wire ran through each line of LEDs, unlike our flimsy first sheet. Although efficient, the Curly-Q method was consuming in both time, and solder wire. Throughout the course of the twenty plus hours we spent soldering, we ran through just over three entire roles of solder wire weighing 500 grams each. While the cube was being built, we also began to build our circuit that would eventually power and control the cube, fan, lcd timer, and off sensor. Using the Arduino RedBoard, we were able to quite efficiently hook up every item on the alarm clock to the power on the RedBoard.

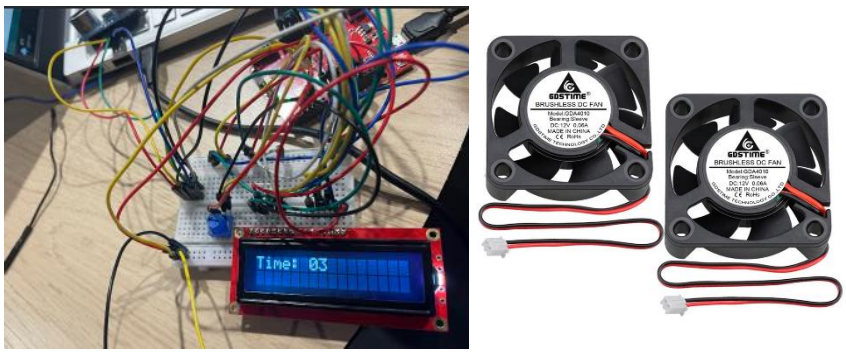

Image 6. Completed circuit, LCD, and fan

For the most part the circuit board worked ok. However, because of all the power the cube required to light up, the board tended to overheat when the power was left on too

long. To fix this we simply let the board cool down and then try again. The process of wiring using the breadboard however is very tedious. Many times, throughout the project we'd accidentally misalign a pin on the circuit board that would cause the circuit to immediately fail

Summary and conclusion:

For our project, we wanted to better the lives of a group of people who already suffer enough in their day-to-day life. The deaf-friendly alarm clock was designed for this sole purpose, as it was built to gently wake the user from their slumber, unlike the abrupt methods of doing so currently on the market such as vibrating pillows or electric shock bands. Creating an alarm clock for deaf individuals is beneficial because it provides them with an effective way to wake up independently. Traditional alarm clocks rely on sound, which isn't useful for those who are deaf or hard of hearing. An alarm clock designed for this community typically utilizes visual or tactile cues, such as flashing lights or vibrations, to alert them when it's time to wake up. This promotes independence, allows for customizable wake-up methods, and ensures that deaf individuals can adhere to their schedules without relying on others. Our final product alarm clock consisted of an 8x8x8 LED cube that displayed multiple captivating light patterns, a fan, digital clock/timer, and an off switch/sensor. Throughout the course of the project, we ran into multiple problems. We soldered incorrectly, wired the circuit wrong on multiple occasions, had bugs in our code, ran out of solder wire, etc. Once the wiring was completed and the code was uploaded, we hid the exposed wire in a cardboard box, while making cutouts for the LCD, power cable, fan, and cube. Yet because of the fragility of the wiring as well, the original box we tried to embed the lcd, fan, and sensor in did not work as it was an extremely tight fit, and wires constantly fell out once inside the box. This set us back quite a bit at rewiring the circuit consumed a lot of time. Because of these issues, a few future suggestions for the next iteration of our alarm clock would be to find a better-quality circuit board and stick the wires in the breadboard more securely. Overall, the project was a success, our problem of assisting the deaf in their daily endeavors is sure to be solved with our innovation.

References

[1] RNID, "Prevalence of deafness and hearing loss," [Online]. Available: https://rnid.org.uk/get-involved/research-and-policy/facts-and-figures/prevalence-of-deafness-and-hearing-loss/. [Accessed: April 18th, 2024].

[2] RNID, "Alarms," [Online]. Available: https://rnid.org.uk/information-and-support/technology-and-products/alarms/. [Accessed: April 18th, 2024].

[3] Start ASL, "Deaf alarm clock," [Online]. Available: https://www.startasl.com/deaf-alarm-clock/. [Accessed: April 18th, 2024].